Power Engineering and Engineering Thermophysics
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Experimental Design and Performance Evaluation of a Solar-Assisted Wearable Cooling Vest With Dual Forced-Air Ventilation

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Received: 07-30-2026 **Revised:** 09-13-2026 **Accepted:** 09-30-2026

Citation: H. L. Tbena, Z. A. Oudah, M. M. Mansour, A. A. Ouda, and A. M. Lafta, "Experimental design and performance evaluation of a solar-assisted wearable cooling vest with dual forced-air ventilation," *Power Eng. Eng. Thermophys.*, vol. 5, no. 4, pp. 282–294, 2026. https://doi.org/10.56578/peet050403.

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Abstract: Lightweight personal thermal-management systems are attractive for outdoor heat exposure where space cooling is unavailable. A low-power, solar-assisted garment with dual forced-air ventilation was fabricated and tested outdoors on a single day (08:00–16:00; n = 9 nominal hourly observations, eight elapsed one-hour intervals) at ambient temperatures of 32.5–43.1 ℃, relative humidity of 20–34%, and solar irradiance of 420–950 W/m^2. The apparent fan-OFF/fan-ON garment-temperature contrast, $\Delta T_g = T_{OFF} - T_{ON}$, was positive at all nine observations, with a mean of 3.89 ± 0.97 ℃ and a maximum of 5.0 ℃ at 14:00. Based on the reported combined airflow and ΔT_g, the estimated sensible cooling rate was $\dot{Q}_{s,est}$ = 38.43–99.52 W (nine-point mean 75.08 W), and the ventilation cooling performance ratio was VCPR = $\dot{Q}_{s,\,est}/P_{fan}$ = 13.72–35.54 (nine-point mean 26.81); these are engineering screening estimates rather than calorimetric measurements or refrigeration coefficient of performance (COP) values. Trapezoidal integration over the eight elapsed one-hour intervals gave a fan-energy demand of 22.40 Wh, a photovoltaic (PV) panels contribution of 13.08 Wh, and an auxiliary-energy requirement of 9.33 Wh. The PV contribution therefore accounted for 58.4% of the fan-energy demand during the test period. Single-day associations with the ambient conditions are reported descriptively, without causal interpretation. The results provide proof-of-concept evidence of a consistent apparent garment-temperature contrast and a quantified partial PV load offset under the hot-dry conditions tested, and they define an 11.19 Wh storage target for autonomous daytime operation.

Keywords: Wearable cooling vest; Forced-air ventilation; Personal thermal management; Photovoltaic assistance; Garment microclimate; Heat stress; Energy balance; Ventilation cooling performance ratio.

1 Introduction

Physical performance, concentration, comfort and occupational safety are compromised by exposure to heat, especially when the task is undertaken outdoors for extended periods in hot, dry climates [1–3]. Personal thermal management systems focus on the microenvironment near the body and can help to decrease reliance on space conditioning. The following methods have been developed: phase change garments, liquid based garments, thermoelectric devices and fan assisted ventilation [4–11].

Liquid and thermoelectric systems can provide active cooling, but they require pumps, tubing, heat exchangers, heat sinks, or a relatively large electrical supply [12–15]. Fan-assisted garments, on the other hand, are mechanically simpler, because low-power fans circulate air within the garment microclimate to aid convective and evaporative heat transfer when ambient conditions are suitable [16–21]. Direct evidence that a small, body-mounted photovoltaic (PV) module can supply part of the electrical load of a wearable system is available from two lines of work: wearable-scale flexible PV panels mounted on the body delivered an average of approximately 65 mW and a peak of approximately 94 mW under outdoor irradiance and were proposed explicitly as a battery-supplementing supply for body-worn electronics [22], and solar-powered thermoregulatory clothing has been demonstrated in which an integrated PV layer drives the thermal-management element of the garment using sunlight alone [23]. Because the electrical power delivered by such a module depends on its electrical operating point and on its own temperature, an explicit treatment of the PV thermal behaviour is required when the module is mounted on a garment exposed to high irradiance [24, 25]; PV sizing studies for small stand-alone direct-current loads provide the corresponding framework for matching the daily PV yield, the load demand, and the required storage capacity [26].

This work analyzes a physically fabricated proof-of-concept garment that combines two separate axial fans with direct PV assistance. This study is limited to direct observations of the garment microclimate and the electrical power balance. It does not claim physiological cooling, improved thermal comfort, or a refrigeration-cycle coefficient of performance (COP).

2 Literature Review and Research Gap

2.1 Existing Wearable-Cooling Approaches

Wearable cooling technologies differ in cooling capacity, duration, mass, complexity, and electricity consumption. Phase-change systems have finite latent-heat storage; liquid-circulation systems provide cooling but require pumps and tubing; thermoelectric systems provide local active heat pumping but require hot-side heat rejection; passive garment systems are highly sensitive to radiative and environmental boundary conditions; and forced-air systems have low electrical demand but are sensitive to airflow distribution and ambient humidity. Table 1 positions the present prototype relative to these approaches.

Table 1. Position of the present prototype relative to major wearable-cooling approaches

Approach	Primary Mechanism	Main Strength	Principal Limitation	Relation to Present Study
Phase-change garment	Latent-heat storage	Quiet; no continuous power during discharge	Finite duration and required regeneration	Potential future hybrid storage
Liquid-circulation garment	Pumped cooled liquid	High and distributed heat removal	Pump, tubing, reservoir, and cooling-source mass	Higher complexity than the present prototype
Thermoelectric garment	Peltier heat pumping	Electrically controlled local cooling	Hot-side heat rejection and battery demand	Not used in the evaluated configuration
Passive/radiative textile	Optical and radiative heat management	Very low electrical demand	Strong dependence on exposure and environment	Possible complementary strategy
Forced-air garment	Air exchange within clothing microclimate	Lightweight and low power	Humidity- and distribution-dependent response	Core cooling mechanism
Present prototype	Dual forced-air ventilation with photovoltaic (PV) electrical assistance	Low fan demand and reduced auxiliary energy	Single-day proof-of-concept; no wearer	Thermal observations plus quantified PV energy balance

2.2 Research Gap and Objective

The feasibility of wearable cooling garments based on vented, phase-change, liquid, thermoelectric, and multimodal technologies has been established in controlled laboratory and workplace studies [8, 9, 13, 27, 28]. However, fewer studies explicitly report a matched thermal record and an explicit daytime electrical balance for a low-complexity, garment-mounted PV/dual-fan configuration at air temperatures above 40 ℃. The objective of this study is therefore to quantify (i) the apparent fan-OFF/fan-ON garment-temperature contrast, (ii) the estimated sensible cooling rate and the associated ventilation cooling performance ratio (VCPR) screening index, and (iii) the fraction of the fan-energy demand supplied by the PV module over the recorded eight-hour period. A further aim is to provide a transparent chain from the archived observations to a storage-sizing target, in which observed, exactly derived, and assumption-based quantities are kept separate (Section 3).

3 Materials and Methods

3.1 Prototype Configuration

The prototype consisted of a lightweight outer garment with a hood, two small axial fans placed at two different locations on the lower torso, a PV panel attached to the outer garment, electrical wiring, and leads for the temperature sensors (Figures 1–4). The combined reported demand of the two fans was 2.80 W at a nominal voltage of 5 V. Part of this load was supplied by the PV panel, and the remainder was supplied by an auxiliary source. The archived dataset reports a combined airflow of 57–62 m^3/h for the dual-fan system. Because the airflow instrument and measurement

plane were not preserved in the experimental archive, this series is used only as an input to the screening calculation defined in Section 3.4.



Figure 1. Fabricated solar-assisted garment showing the photovoltaic (PV) panel and the two ventilation fans



Figure 2. Close view of one integrated axial ventilation fan

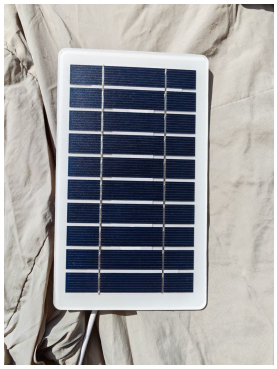


Figure 3. Close view of the garment-mounted photovoltaic (PV) panel



Figure 4. Inner side of the experimental garment showing electrical wiring and measurement leads

3.2 Instrumentation and Data Acquisition

The data provenance was checked against the prototype photographs and the archived nine-row time series. The retained archive does not include instrument nameplates, manufacturer/model identifiers, calibration certificates, accuracy classes, or original acquisition files. Table 2 therefore distinguishes between the information preserved in the archive and the metrological metadata that were not preserved. Where the supporting record is not available, no numerical uncertainty, calibration status, or instrument accuracy is claimed.

Table 2. Measurement provenance, archived metadata status, and permitted use

Quantity	Archived Numerical Record	Metrological Metadata Status	Permitted Analytical Use
Ambient and garment temperatures	Ambient: 32.5–43.1 °C; paired fan-OFF/fan-ON garment-temperature series	Sensor leads are visible in Figure 4. Manufacturer/model, sensor type, range, resolution, accuracy, calibration record, and exact sensing coordinates were not preserved.	Reported directly as archived observations; no propagated temperature uncertainty or physiological interpretation.
Relative humidity	20–34% at the nine nominal hourly records	Instrument identity, range, resolution, accuracy, shielding, height, and calibration record were not preserved.	Environmental context and descriptive co-variation only; no independent causal effect is inferred.
Solar irradiance	420–950 W/m^2 at the nine nominal hourly records	Meter identity, range, resolution, accuracy, calibration record, and measurement plane relative to the PV surface were not preserved.	Exposure description and descriptive association with reported PV power only.
Combined airflow	57–62 m^3/h, reported for the dual-fan system	Airflow-instrument identity, probe plane, area-averaging procedure, individual-fan values, resolution, accuracy, and calibration record were not preserved.	Input to $\dot{Q}_{s,est}$ as a screening calculation only; not treated as traceable duct-flow measurement.
Electrical power	Fan demand: 2.80 W; photovoltaic (PV) output: 0.85–2.10 W	The archive does not preserve whether power was read directly or reconstructed from synchronized voltage and current; meter identities and calibration records were not preserved.	The reported power series is integrated directly for the internal energy balance; no instrument-accuracy claim is made.
Time base	08:00–16:00; nine tabulated points at 1 h spacing	The internal logger sampling rate, if any, was not preserved.	Eight elapsed one-hour intervals are used for trapezoidal integration.

This provenance boundary is maintained throughout the analysis: the values are not reported as calibrated measurements, and no numerical uncertainty budget is constructed retrospectively. Derived quantities are therefore provided either as exact arithmetic transformations of the archived series or, for $\dot{Q}_{s,est}$ and VCPR, as explicit assumption-based screening estimates. Figure 5 summarizes this three-tier structure, which is maintained in all subsequent tables and figures.

3.3 Outdoor Test Protocol and Comparison Basis

The archived photographs show the garment without a wearer or a thermal manikin (Figures 1–4). The reported temperatures therefore describe an unworn garment microclimate and should not be interpreted as skin temperature, physiological strain, or perceived comfort. The single daytime run provided nine nominal one-hour time points and eight elapsed one-hour intervals.

Each tabulated record contains a fan-ON value and a fan-OFF value at each nominal time. The retained record does not establish whether the fan-ON and fan-OFF values were obtained simultaneously, successively, or in an alternating sequence, nor whether the same garment specimen was used for both operating states. No separate stabilization interval was recorded. Hence, the operating-state contrasts are reported only as within-pair contrasts tabulated for the same nominal time; they are not described as randomized, independent, steady-state, or unbiased treatment effects.

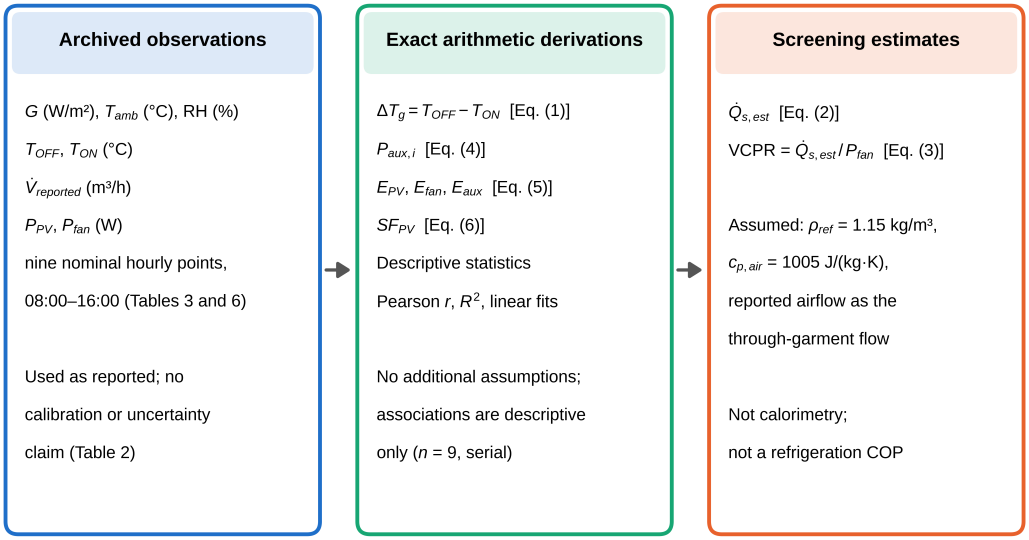


Figure 5. Analysis framework separating the archived observations, the exact arithmetic derivations, and the assumption-based screening estimates used in this study

No replicate-run dataset is available; the archive contains a single test day. No compass azimuth or quantitative panel-tilt record is available. The photographs show the PV module suspended vertically and facing the sun at the time the photographs were taken. Accordingly, no orientation-specific claim is made, and outdoor variability, possible state-order effects, and the lack of independent replication are treated as design constraints.

The two operating states and the nine consecutive readings are not assumed to be statistically independent, because the outdoor temperature, humidity, irradiance, and time varied together. Comparisons are therefore limited to the recorded operating range and are described as single-day exploratory observations.

3.4 Apparent Garment-Temperature Contrast, Estimated Sensible Cooling Rate, and Ventilation Cooling Performance Ratio

The apparent fan-OFF/fan-ON garment-temperature contrast is defined as the difference between the garment microclimate temperatures recorded in the two operating states:

$$\Delta T_{\mathrm{g}} = T_{\mathrm{OFF}} - T_{\mathrm{ON}} \tag{1}$$

where, ΔT_{g} is an operating-state contrast rather than a measured inlet-to-outlet air-temperature change across the fans. It therefore cannot be presented as direct calorimetry and is used only in the following explicitly labelled screening estimate of the sensible cooling rate:

$$\dot{Q}_{\mathrm{s,est}} = \rho_{\mathrm{ref}} c_{\mathrm{p,air}} \dot{V} \Delta T_{\mathrm{g}}, \dot{V} = \frac{\dot{V}_{\mathrm{reported}}}{3600} \tag{2}$$

where, ρ_{ref} =1.15 kg/m^3 is a reference density representative of hot ambient air, $c_{\mathrm{p,air}}$ = 1005 J/(kg·K) is the specific heat of air, and $\dot{V}$ is the reported combined airflow, $\dot{V}_{\mathrm{reported}}$ (m^3/h), converted to m^3/s. Because the archive does not preserve the airflow instrument or the measurement plane, $\dot{Q}_{\mathrm{s,est}}$ is an estimated (screening) sensible cooling rate based on the reported airflow and the fan-OFF/fan-ON temperature contrast; it is not a measured heat-removal rate and is not used as an energy-conservation closure. The sensitivity of $\dot{Q}_{\mathrm{s,est}}$ to the density assumption is examined in Section 4.2.

The dimensionless VCPR is defined consistently as:

$$\mathrm{VCPR} = \frac{\dot{Q}_{\mathrm{s,est}}}{P_{\mathrm{fan}}} \tag{3}$$

VCPR expresses the estimated sensible cooling rate per unit of fan electrical power under the screening convention of Eq. (2). It is not a COP of a vapor-compression or thermoelectric refrigeration cycle. Accordingly, no performance value is reported with refrigeration COP or COP$_{\mathrm{eff}}$ designation, and VCPR is used only under this definition.

3.5 Electrical Power Balance and Auxiliary Energy Requirement

The analysis therefore uses the archived power values exactly as reported and makes no claim about the acquisition path. The auxiliary power requirement at each observation time is:

$$P_{\text{aux},i} = \max\left(0, P_{\text{fan},i} - P_{\text{PV},i}\right) \tag{4}$$

The nine power values span eight elapsed one-hour intervals. Energy was therefore obtained by trapezoidal integration rather than by multiplying the number of observations by one hour; the latter would implicitly assign nine hours to an eight-hour record and would overstate the fan-energy demand by 12.5% (25.20 Wh instead of 22.40 Wh):

$$E = \sum_{i=1}^{n-1} \frac{P_i + P_{i+1}}{2} \left(t_{i+1} - t_i\right), n = 9 \tag{5}$$

For the recorded PV series, E_{PV} =13.075 Wh ≈ 13.08 Wh. Because the fan demand was constant at 2.80 W over the eight elapsed hours, the fan-energy demand was E_{fan} = 2.80 × 8 h = 22.40 Wh. PV power was lower than the fan demand at every observation; hence, E_{aux} =E_{fan} − E_{PV} = 9.325 Wh ≈ 9.33 Wh. The PV contribution fraction (solar fraction) over the test period was:

$$SF_{\text{PV}} = \frac{E_{\text{PV}}}{E_{\text{fan}}} \times 100 = \frac{13.075}{22.40} \times 100 = 58.37\% \approx 58.4\% \tag{6}$$

3.6 Exploratory Statistical Analysis

Pearson correlation coefficients and simple linear fits were computed for selected variable pairs using the nine consecutive hourly observations recorded on the single test day $(n = 9)$. Because these observations are few and serially correlated through time-varying weather conditions, formal hypothesis testing is not applied, and the nine rows are not treated as independent replicates. To document this dependence, the inter-correlations among the environmental variables, the lag-1 autocorrelation of ΔT_{g}, and the Durbin-Watson statistic of the $\Delta T_{\text{g}} - T_{\text{amb}}$ fit residuals are also reported. Accordingly, r, R^2, and the fitted slopes describe only the observed within-day co-variation; they do not identify causal effects and do not provide a predictive model beyond the recorded range.

4 Results and Discussion

4.1 Garment-Temperature Response Under Outdoor Conditions

Ambient temperature rose from 32.5 ℃ at 08:00 to 43.1 ℃ at 14:00, irradiance peaked at 950 W/m^2 at 13:00, and the relative humidity decreased from 34% at 08:00 to 20% at 14:00 (Table 3 and Figure 6). The garment temperature recorded in the fan-ON state was lower than that recorded in the fan-OFF state at every nominal time (Figure 7). The apparent fan-OFF/fan-ON garment-temperature contrast, ΔT_{g}, ranged from 2.1 ℃ at 08:00 to a maximum of 5.0 ℃ at 14:00 and averaged 3.89 ℃ across the nine time points (Figure 8).

Table 3. Recorded environmental, garment-temperature, and airflow values

Time	G (W/m^2)	T_{amb} (℃)	RH (%)	T_{OFF} (℃)	T_{ON} (℃)	ΔT_{g} (℃)	Airflow (m^3/h)
08:00	420	32.5	34	34.1	32.0	2.1	57
09:00	560	34.2	31	36.2	33.4	2.8	58
10:00	700	36.1	28	38.5	35.1	3.4	59
11:00	820	38.4	25	40.8	36.8	4.0	60
12:00	910	40.7	23	42.9	38.4	4.5	61
13:00	950	42.3	21	44.1	39.3	4.8	61
14:00	900	43.1	20	44.8	39.8	5.0	62
15:00	760	42.0	21	43.7	39.1	4.6	61
16:00	580	40.1	24	41.9	38.1	3.8	59

Note: G = Solar irradiance; T_{amb} = Ambient air temperature; RH = Relative humidity; T_{OFF}, T_{ON} = Garment temperature under fan-OFF/ON conditions; ΔT_{g} = difference of temperature.

The descriptive summary of ΔT_{g} was 3.89 ± 0.97 ℃ (mean ± sample standard deviation), with a median of 4.0 ℃, an interquartile range of 3.4–4.6 ℃, and an eight-hour trapezoidal integral of 32.05 ℃·h. From 10:00 to 16:00, ΔT_{g} remained at or above 3.4 ℃.

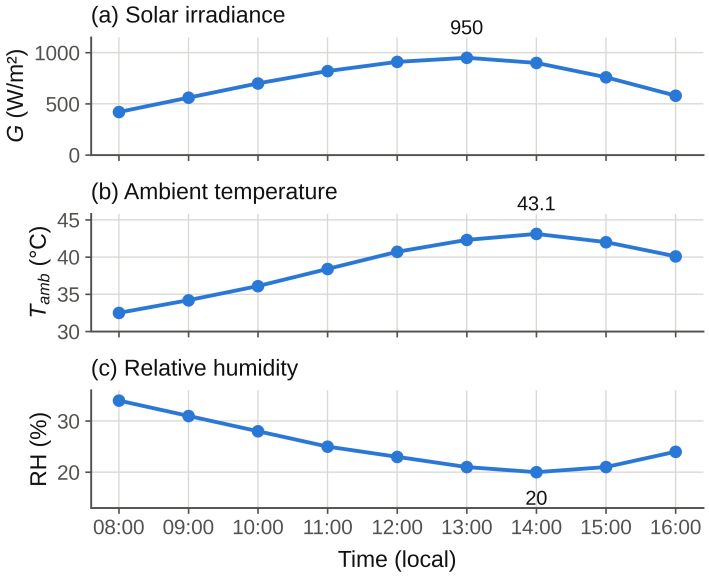


Figure 6. Recorded environmental conditions during the outdoor test: (a) solar irradiance, G; (b) ambient temperature, T_{amb}; (c) relative humidity, RH

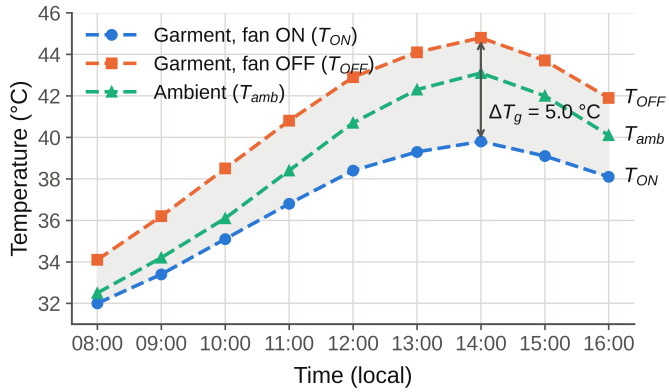


Figure 7. Hourly ambient and garment microclimate temperatures under fan-OFF and fan-ON conditions

Note: The shaded band represents the apparent garment-temperature contrast, ΔT_{g}.

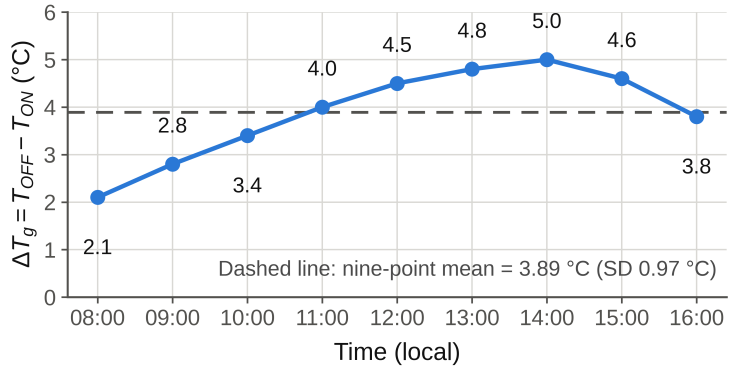


Figure 8. Apparent fan-OFF/fan-ON garment-temperature contrast, $\Delta T_{\mathrm{g}} = T_{\mathrm{OFF}} - T_{\mathrm{ON}}$, at the nine nominal observation times

Note: The dashed line marks the nine-point mean (3.89 °C).

4.2 Estimated Sensible Cooling Rate and Ventilation Cooling Performance Ratio

Using the reported combined airflow and the screening relation in Eq. (2), the estimated sensible cooling rate, $\dot{Q}_{\mathrm{s,est}}$, ranged from 38.43 to 99.52 W, with a nine-point arithmetic mean of 75.08 W (Table 4 and Figure 9). The

calculation used the reported total airflow converted from m^3/h to m^3/s, ρ_{ref} = 1.15 kg/m^3, $c_{p,air}$ = 1005 J/(kg·K), and the apparent garment-temperature contrast $\Delta T_g = T_{OFF} - T_{ON}$.

Table 4. Estimated (screening) sensible cooling rate, $\dot{Q}_{s,est}$ and ventilation cooling performance ratio (VCPR) at the nine nominal observation times

Time	$V_{reported}$ (m^3/h)	ΔT_g (°C)	$\dot{Q}_{est}$ (W)	VCPR
08:00	57	2.1	38.43	13.72
09:00	58	2.8	52.14	18.62
10:00	59	3.4	64.40	23.00
11:00	60	4.0	77.05	27.52
12:00	61	4.5	88.13	31.47
13:00	61	4.8	94.00	33.57
14:00	62	5.0	99.52	35.54
15:00	61	4.6	90.08	32.17
16:00	59	3.8	71.98	25.71
Mean ($n = 9$)	59.78	3.89	75.08	26.81

Note: $V_{reported}$ (m^3/h) = the reported combined airflow; ΔT_g = difference of temperature.

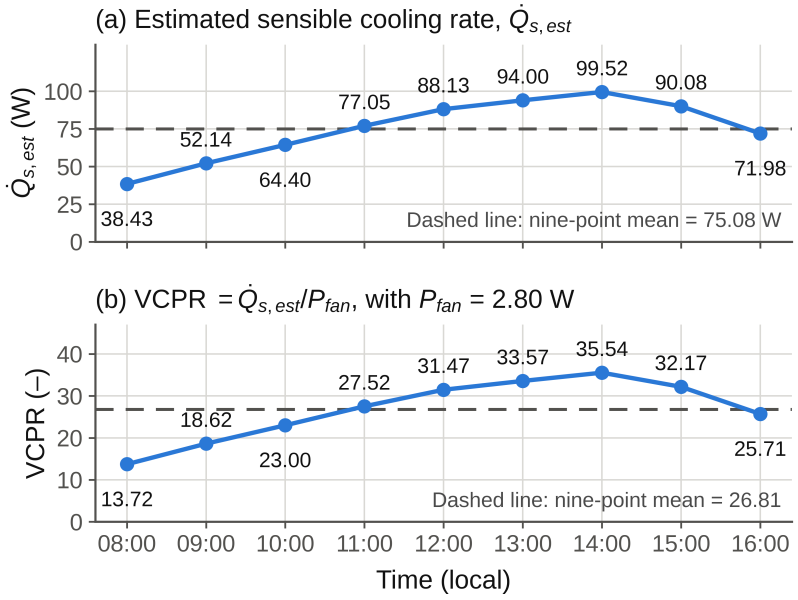


Figure 9. At the nine nominal observation times: (a) estimated sensible cooling rate, $\dot{Q}_{s,est}$; (b) VCPR = $\dot{Q}_{s,est}/P_{fan}$
Note: VCPR = ventilation cooling performance ratio; P_{fan} = fan electrical powers. Dashed lines mark the nine-point means (75.08 W and 26.81). Both quantities are screening indices based on the reported airflow and the apparent fan-OFF/fan-ON garment-temperature contrast.

Because ΔT_g does not represent a directly measured inlet-to-outlet air-temperature difference and the archived record does not preserve the airflow measurement plane or instrument metadata, $\dot{Q}_{s,est}$ should be interpreted only as an estimated (screening) sensible cooling rate based on the reported airflow and the fan-OFF/fan-ON temperature contrast. It is not a direct calorimetric measurement and is not used to establish an energy-conservation closure.

Using the constant reported fan demand of 2.80 W, the corresponding VCPR ranged from 13.72 to 35.54, with a nine-point mean of 26.81. The two means are related by Eq. (3), i.e., 75.08 W/2.80 W = 26.81. The variation of both $\dot{Q}_{s,est}$ and VCPR during the observation period follows arithmetically from Eqs. (2) and (3), which combine the reported airflow and ΔT_g; it should therefore not be interpreted as independent experimental confirmation of enhanced cooling performance.

Because ρ_{ref} is fixed at 1.15 kg/m^3, the calculation was repeated with the hourly dry-air density evaluated at the recorded ambient temperature. This changed the hourly $\dot{Q}_{s,est}$ values by between +0.4% and -2.9%, giving a nine-point mean of 73.67 W (-1.9%) and a mean VCPR of 26.31. The density assumption therefore has only a minor influence, whereas the airflow assignment remains the dominant, unquantified source of uncertainty in the screening estimates.

4.3 Exploratory Daily Associations

Over the nine consecutive observations, ΔT_g co-varied positively with ambient temperature (r = 0.978, R^2 = 0.957) and irradiance (r = 0.912, R^2 = 0.832), and negatively with relative humidity (r = -0.988, R^2 = 0.976) (Table 5). PV output increased approximately linearly with irradiance (r = 0.998, R^2 = 0.996). These coefficients describe shared single-day trajectories; they do not attribute the variation in ΔT_g to any individual environmental variable.

Table 5. Exploratory associations for the single-day time series ($n = 9$ consecutive observations)

Variable Pair	n	Pearson r	R^2	Interpretive Scope
Ambient temperature vs. ΔT_g	9	0.978	0.957	Exploratory association
Solar irradiance vs. ΔT_g	9	0.912	0.832	Exploratory association
Relative humidity vs. ΔT_g	9	-0.988	0.976	Exploratory association
Solar irradiance vs. photovoltaic (PV) output	9	0.998	0.996	Exploratory association

Note: ΔT_g = difference of temperature.

The corresponding linear fits, valid only within the recorded ranges, are ΔT_g = 0.252T_{amb} – 5.884, ΔT_g = 0.00486G + 0.323, ΔT_g = -0.199RH + 8.902, and P_{PV} = 0.00236G – 0.165 (dashed lines in Figure 10). They are descriptive single-day fits and should not be used for prediction, extrapolation, or causal attribution.

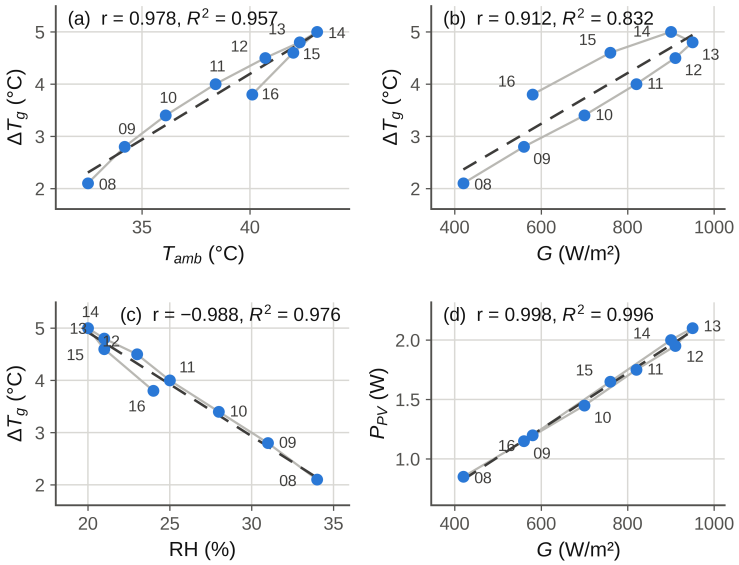


Figure 10. Exploratory single-day associations: ΔT_g versus (a) ambient temperature; (b) solar irradiance; (c) relative humidity; (d) photovoltaic (PV) output versus solar irradiance

Note: Points are joined in time order and labelled by hour; dashed lines are descriptive least-squares fits within the recorded range.

The environmental variables were themselves strongly inter-correlated over the day (T_{amb}–RH, r = -0.995; T_{amb}–G, r = 0.818; G–RH, r = -0.843), and ΔT_g was serially dependent (lag-1 autocorrelation = 0.63; Durbin–Watson statistic of the ΔT_g–T_{amb} fit residuals = 0.68). Figure 10 also shows that the afternoon observations did not retrace the morning ones: at similar ambient temperatures, ΔT_g was 4.5 °C at 12:00 (40.7 °C) but 3.8 °C at 16:00 (40.1 °C), and the ΔT_g–G relation forms a loop rather than a single line. The individual associations therefore cannot be separated from one another or from the time of day, and they are reported descriptively only.

4.4 Eight-Hour Energy Balance and Photovoltaic Output

PV output increased from 0.85 W at 08:00 to a maximum of 2.10 W at 13:00, which coincided with the peak irradiance of 950 W/m^2 (Table 6 and Figure 11). The maximum PV output corresponded to 75% of the instantaneous fan demand of 2.80 W, with the remaining 0.70 W supplied by the auxiliary source. The hourly values in Table 6 are instantaneous powers (W), whereas the integrated quantities are energies (Wh). The energy balance was obtained by trapezoidal integration over the eight one-hour intervals between the nine observations (Eq. (5)); it gives a fan-energy

demand of 22.40 Wh, a PV contribution of 13.08 Wh, and an auxiliary-energy requirement of 9.33 Wh. The PV contribution therefore accounted for 58.4% of the fan-energy demand (Figure 11).

Table 6. Hourly electrical powers (W) and integrated eight-hour energy balance (Wh)

Time	Photovoltaic (PV) Output (W)	Fan Demand (W)	Auxiliary Requirement (W)
08:00	0.85	2.80	1.95
09:00	1.15	2.80	1.65
10:00	1.45	2.80	1.35
11:00	1.75	2.80	1.05
12:00	1.95	2.80	0.85
13:00	2.10	2.80	0.70
14:00	2.00	2.80	0.80
15:00	1.65	2.80	1.15
16:00	1.20	2.80	1.60
Integrated energy, 08:00–16:00	13.08 Wh	22.40 Wh	9.33 Wh

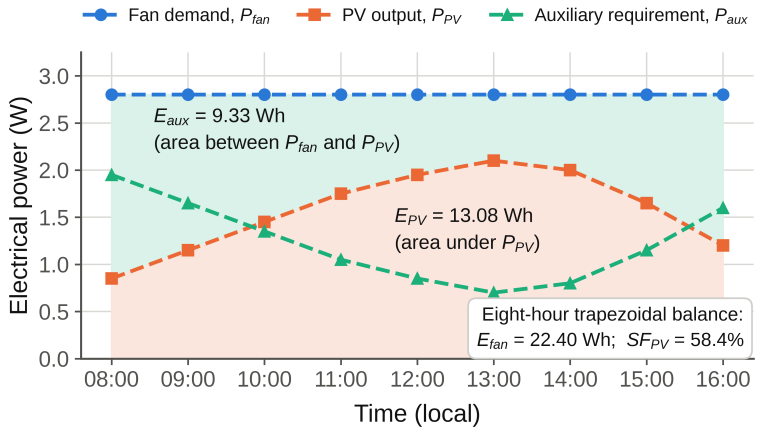


Figure 11. Hourly fan demand, photovoltaic (PV) output, and auxiliary electrical requirement (W)

Note: The shaded areas represent the integrated eight-hour energies: E_{PV} = 13.08 Wh (area under the PV-output curve) and E_{aux} = 9.33 Wh (area between the fan-demand and PV-output curves), giving E_{fan} = 22.40 Wh and SF_{PV} = 58.4%.

4.5 Combined Interpretation

The largest apparent garment-temperature contrast (4.8–5.0 °C at 13:00–14:00) coincided with high ambient temperature and high irradiance, when the PV module supplied 2.0–2.1 W. The maxima of $\dot{Q}_{\mathrm{s,est}}$ and VCPR also occurred at 14:00; this coincidence is arithmetic, because both indices are calculated from ΔT_{g} and the reported airflow, and it is not presented as independent confirmation. The single-day record does not allow the influences of ambient temperature, humidity, irradiance, time of day, garment state, and any unrecorded change in orientation or wind to be separated. The findings therefore represent an internally consistent operating-state contrast and electrical balance, not an environmental cause-and-effect relationship.

For context, the reported combined airflow of 57–62 m^3/h (15.8–17.2 L/s) is of the same order as the 12 L/s fan ventilation [29], in which fan ventilation increased total torso cooling by 137–251%. Because the present garment was unworn and the airflow measurement plane was not documented, these values are not directly comparable; the comparison indicates only that the airflow level of the prototype lies within the range examined in controlled ventilation-garment studies.

5 Practical Implications and Design Improvements

This eight-hour balance can be directly converted into the storage requirement. So the daytime auxiliary-energy deficit is 9.325 Wh and the engineering reserve is 20%, meaning that the minimum usable storage is 11.19 Wh. Prior to converter, wire, ageing and temperature derating, a capacity of 2.24 Ah at 5 V or 0.93 Ah at 12 V. These

nominal values, therefore, need to be surpassed by the selected battery capacity by considering the actual conversion efficiency and the allowable depth of discharge. This sizing result is more descriptive than a statement of full solar autonomy: the PV source was able to provide 58.4% of the fan energy demand for an 8-hour period, but was unable to provide the entire 2.80 W demand for any time period. A design well suited for the field should thus incorporate a charge controller, an energy buffer, protected wiring, washable detachable electronics, acoustic and garment-mass measurements, and repeatable through-garment airflow mapping. To advance from proof of concept, future experiments should feature matched exposure, fixed orientation, a prespecified stabilization criterion, repeated exposures on separate days, ambient and garment temperature sensors being shielded and co-located, and measured inlet–outlet air temperatures. Any reference to skin temperature, comfort, physiological strain or occupational protection should always be followed by a thermal-manikin or ethics approved wearer protocol.

6 Limitations

This study consists of nine replications (observations) that are serially ordered. No records of fan-OFF/fan-ON sequencing, stabilization period, garment azimuth, PV tilt or repeat runs are available. The parameters listed above are not included in the archived files, which include the instrument manufacturer, model, range, resolution, stated accuracy, calibration/reference checks, exact sensor coordinates, the plane of airflow sampling, and the path of electrical power acquisition.

These items are therefore reported as unavailable metadata rather than replaced by assumed specifications, and no metrological uncertainty or calibrated-performance claim is made. The garment was not evaluated with a wearer, and no conclusions are drawn regarding skin temperature, thermal sensation, comfort, physiological strain, or occupational safety. The apparent fan-OFF/fan-ON garment-temperature contrast is not an inlet-to-outlet air-temperature difference. Therefore, $\dot{Q}_{s,est}$ and VCPR are screening indices based on assumed air properties and the reported airflow; they are not direct calorimetric measurements and should not be regarded as refrigeration COP values. No performance value is reported as COP or COP_{eff}. The radiation shielding and exposure of the ambient and garment sensors were also not documented: T_{ON} was 0.5–3.3 °C below the recorded T_{amb}, whereas T_{OFF} was 1.6–2.4 °C above it. Because such cross-series differences may reflect different sensor exposure, only the within-garment contrast ΔT_g is interpreted, and garment-to-ambient differences are not used in any derived quantity. Finally, the single-day correlations are confounded by time and by co-varying weather variables; they cannot establish causal effects or support prediction beyond the observed ranges.

7 Conclusions

A dual-fan, PV-assisted garment prototype was tested outdoors during a single hot-dry day from 08:00 to 16:00. The apparent fan-OFF/fan-ON garment-temperature contrast, ΔT_g, was positive at all nine observations, with a mean of 3.89 ± 0.97 °C and a maximum of 5.0 °C at 14:00. Under the stated screening assumptions, the estimated sensible cooling rate, $\dot{Q}_{s,est}$, ranged from 38.43 to 99.52 W, while VCPR ranged from 13.72 to 35.54 (nine-point means of 75.08 W and 26.81, respectively). These values are engineering screening indices rather than direct calorimetric measurements or refrigeration COP values; replacing the fixed air density with a temperature-dependent value changed the mean $\dot{Q}_{s,est}$ by only -1.9%. The eight-hour trapezoidal electrical balance gave a fan-energy demand of 22.40 Wh, a PV contribution of 13.08 Wh, and an auxiliary-energy requirement of 9.33 Wh, corresponding to a PV contribution of 58.4%. The corresponding usable energy-buffer target is 11.19 Wh, including a 20% reserve. Results appear to show a garment-temperature contrast and the partial PV load offset, but not a measured rate of heat removal or an environmental cause-and-effect relationship. Before the results can be generalized, traceable, replicated, and counterbalanced tests are required.

Author Contributions

Conceptualization, H.L.T., Z.A.O., and M.M.M.; methodology, H.L.T. and M.M.M.; software, H.L.T.; validation, H.L.T., Z.A.O., and A.A.O.; formal analysis, H.L.T. and A.M.L.; investigation, H.L.T. and Z.A.O.; resources, M.M.M. and A.A.O.; data curation, H.L.T.; writing—original draft preparation, H.L.T.; writing—review and editing, M.M.M., Z.A.O., and A.M.L.; visualization, H.L.T.; supervision, M.M.M.; project administration, M.M.M.; funding acquisition, M.M.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

The complete nine-point dataset analysed in this study (environmental conditions, garment temperatures, airflow, and electrical powers) is reported in Tables 3, 4, and 6. The additional data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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Nomenclature

$c_{\text{p,air}}$	Specific heat of air, J/(kg·K)
E	Energy integrated over the test period, Wh
E_{aux}, E_{PV}, E_{fan}	Auxiliary, PV, and fan energies, Wh
G	Solar irradiance, W/m^2
n	Number of observations (n = 9)
P_{aux}, P_{PV}, P_{fan}	Auxiliary, PV, and fan electrical powers, W
$\dot{Q}_{\text{s,est}}$	Estimated (screening) sensible cooling rate, W
r, R^2	Pearson correlation coefficient; coefficient of determination
RH	Relative humidity, %
SF_{PV}	PV contribution fraction (solar fraction), %
t	Time, h
T_{amb}	Ambient air temperature, ℃
T_{OFF}, T_{ON}	Garment microclimate temperature, fan OFF/fan ON, ℃
$\dot{V}_{\text{reported}}$, $\dot{V}$	Reported combined airflow; the same airflow in SI units, m^3/h; m^3/s
ΔT_{g}	Apparent fan-OFF/fan-ON garment-temperature contrast, ℃
ρ_{ref}	Reference air density, kg/m^3
COP	Coefficient of performance (not used for any result)
PV	Photovoltaic
VCPR	Ventilation cooling performance ratio, $\dot{Q}_{\text{s,est}}/P_{\text{fan}}$